Phase-Adaptive Generation: Learning Block Budgets to Reduce Diffusion-LM Calls

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Abstract

Diffusion language models can revise several tokens per forward call, but their practical efficiency depends on deciding how large a span to attempt and when to stop refining it. Phase-Adaptive Generation (PAG) learns this refinement budget from prior block trajectories while retaining reactive delimiter boundaries. On LLaDA-8B-Instruct, the best corrected PAG result uses 6.7% fewer model calls on MATH-500 with a 1.33-point accuracy decrease; on GSM8K it saves 4.0% with a 1.60-point decrease, while a block-size lookup remains stronger. We develop this result through a sequence of increasingly controlled approaches: token-level phase analysis, categorical/ordinal prediction from 138,901 AdaBlock tuples, cross-model risk calibration, conservative budget ledgers, and verified macro-steps. The cross-model policy reaches a 9.6% normalized call reduction but fails its accuracy gate. Output-preserving policies save at most 1.24%, and batched verification breaks trajectory equivalence on 22 of 64 pilot prompts despite larger apparent call reductions. Together, the experiments show where learned scheduling helps and where it must yield control: histories can allocate refinement capacity, but token commitment requires current-state boundaries and an audited, batch-invariant reference transition.

1 Introduction

Diffusion language models generate by repeatedly revising masked positions, so one model call can advance several tokens at once. This parallelism creates a control problem absent from left-to-right decoding: how large a span should the model attempt, and how long should it refine that span before moving on? Existing decoders answer with fixed schedules or current-step confidence signals [Wu et al., 2026, Lu et al., 2026, Mohamed et al., 2025]. Both are useful, but neither asks whether the trajectory itself forecasts where later compute will be needed.

Phase-adaptive generation (PAG) starts from a simple observation: reasoning traces are not uniformly difficult. Template text and delimiters often stabilize early, while arithmetic, variable binding, and answer finalization remain uncertain for longer. The difficult part is turning this structure into a decoder. Offline accuracy is not enough. The scheduler changes the trajectory that produces its own future features; block sizes are strongly multimodal; and a premature commitment can alter every later block. A useful predictor can therefore be a poor controller.

We develop the method from trace analysis through deployment. We first instrument Dream and LLaDA [Ye et al., 2025, Nie et al., 2025], measure token stabilization, and use change-point detection (CPD) to expose local shifts in uncertainty. Because those boundaries depend strongly on the detector, we then move to 138,901 block-native tuples collected from AdaBlock on 5,000 GSM8K

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training problems. A categorical/ordinal Transformer predicts the next block control from recent history, while the deployed scheduler preserves AdaBlock's delimiter boundary and uses the prediction only to set a refinement budget. Each subsequent approach addresses a failure exposed by the previous one: cross-model calibration tests transfer, risk ledgers protect outputs, and verified macro-steps try to recover useful savings without granting the predictor acceptance authority.

Across the experiments, corrected forward-pass accounting shows that PAG reduces the mean number of forward evaluations (NFE) on LLaDA by 4.0% on GSM8K and 6.7% on MATH-500, with accuracy decreases of 1.60 and 1.33 percentage points. A size-conditioned lookup table is at least as competitive on GSM8K. A later cross-model policy lowers normalized NFE by 9.6% overall, but fails the predeclared accuracy-noninferiority gate because Dream loses substantially more accuracy. Conservative risk ledgers preserve outputs but save less than 1.3%; batched macro-verification appears promising in physical NFE yet fails exact trajectory audits. The same limitation recurs across these protocols: phase histories can forecast computational effort, but they should not decide token acceptance without verification.

Contributions. We make three empirical contributions. First, we connect token-level phase diagnostics to a block-native prediction problem and document why continuous boundary regression fails on a multimodal control space. Second, we give an audited evaluation of learned block-history scheduling, including a stronger history-free control and corrected NFE accounting. Third, we trace seven artifact-backed protocols across two model families, showing how risk calibration, conservative routing, and verified decoding change the efficiency–quality frontier. The outcome is a concrete rule for adaptive dLM decoding: learn capacity allocation, retain reactive semantic boundaries, and keep commitment under an auditable token-level authority.

2 Related work

Masked and block diffusion language models. Discrete and masked diffusion models replace left-to-right factorization with iterative denoising [Lou et al., 2024, Sahoo et al., 2024]. LLaDA and Dream scale this formulation to instruction following and reasoning [Nie et al., 2025, Ye et al., 2025]. Block diffusion introduces a semi-autoregressive interface: the decoder completes one contiguous block before exposing it as context to the next [Arriola et al., 2025]. That interface makes block size and within-block refinement explicit controls. Our work studies whether realized block histories help set those controls; it does not change the model or training objective.

Adaptive and early-exit decoding. Fast-dLLM combines confidence transfer with KV caching and parallel decoding [Wu et al., 2026]. AdaBlock chooses semantic boundaries from delimiter confidence and adapts refinement to the active block [Lu et al., 2026]; it is our native decoder and strongest same-stack baseline. Adaptive Parallel Decoding (APD) learns token-wise denoising paths, whereas SchED and more recent stability rules stop from inter-step similarity or persistent predictions [Israel et al., 2025, Mohamed et al., 2025, 2026]. These methods establish that current-state confidence and stability are useful. PAG asks a different question: whether previous blocks add predictive value after controlling for a strong current-state or size-conditioned rule.

Risk control and verified acceleration. Learn-then-Test and conformal risk control provide finite-sample tools for selecting a policy under a bounded loss [Angelopoulos et al., 2021, 2022]. We apply this machinery to a same-state premature-commitment loss, while evaluating task correctness separately. Self-speculative dLM decoders instead batch draft states and verify them against a reference transition [Gao et al., 2025, Han et al., 2026]. Our later protocols combine these ideas but also audit a systems assumption that is easy to omit: a nominally deterministic transition must remain invariant when evaluated alone or as one row of a batch. What remains unclear in this literature is whether a predictive phase signal survives both closed-loop deployment and the batch implementation of verification. Our contribution is the resulting end-to-end evidence boundary, not a claim of superiority over these contemporaneous decoders.

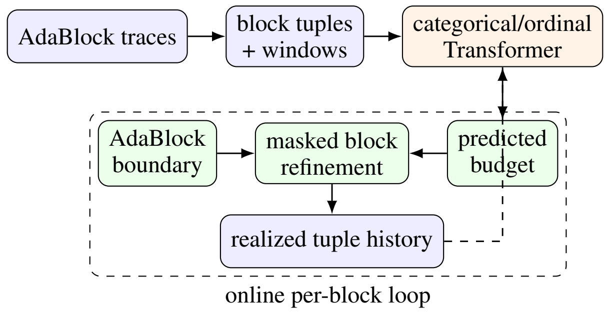
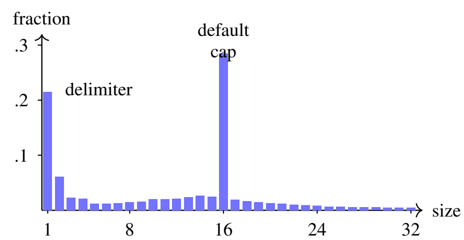


Figure 1: PAG separates two control signals. Current logits determine the next semantic boundary; history predicts a refinement budget. The realized tuple closes the loop only after the block is decoded.

Span	Text excerpt	Mean p
$[0, 2)$	`To solve`	.791
$[2, 22)$	`the equation...`	.992
$[24, 27)$	`will follow`	.669
$[38, 40)$	`parentheses`	.401

(a) One PELT segmentation.



(b) Training block-size distribution.

Figure 2: From token-level phases to block-native data. (a) One algebra trace contains adjacent spans with markedly different stabilization confidence; the exact boundaries vary with detector settings. (b) The 138,901 training blocks form a discrete, multimodal distribution. Size 1 is dominated by delimiter blocks, while size 16 is the default content cap.

3 Phase-adaptive generation

PAG has three stages (Figure 1). Offline trace analysis tests whether the trajectory contains localized compute signals. Block-native traces then supervise a next-control predictor. Online, the predictor shares control with the native decoder: AdaBlock selects the semantic boundary from current logits, while PAG forecasts only the refinement budget. This division was fixed before the audited evaluation.

3.1 Phase signals are diagnostic, not labels

For each output position, we record its predicted identity, top-one and top-two probabilities, and entropy after every refinement call. The stabilization time of token i is the earliest step after which its identity never changes,

$$s_i = \min\{t : \hat{x}_i^{(u)} = \hat{x}_i^{(T)} \text{ for all } u \geq t\}. \tag{1}$$

We read confidence, margin, entropy, and step index at the stabilization time in Equation 1, then segment each feature along token position with Pruned Exact Linear Time (PELT) [Killick et al., 2012]. Figure 2 shows the recurring pattern: long high-confidence spans are interrupted by lower-confidence computation, and block sizes concentrate on delimiter-only and default-cap modes.

The segmentation is intentionally not used as supervision. Boundaries move when the feature, smoothing window, cost, or penalty changes; representative traces yield between two and eight segments under reasonable settings. CPD therefore establishes that trajectories are structured, but does not define a stable scheduler target. The training path instead uses decisions directly observed from a block decoder.

3.2 Block-native next-control prediction

We run AdaBlock with LLaDA-8B-Instruct on 5,000 GSM8K training problems [Cobbe et al., 2021]. After each decoded block, we store its size and realized NFE, stabilization/refinement statistics, mean and minimum confidence, digit fraction, delimiter fraction, and gap statistics. The corpus contains 138,901 training tuples and 5,543 validation tuples. Sliding windows of eight realized tuples predict the next pair $y_t = (b_t, r_t)$, where b_t is block size and r_t is the refinement budget.

The target distribution rules out ordinary regression. A squared-error predictor averages the size-1 delimiter mode and the content-block mode, producing boundaries that seldom occur in the trace corpus. We instead use a Transformer encoder [Vaswani et al., 2017] with a 128-way categorical head for b_t and an 83-threshold ordinal head for r_t. The budget head is conditioned on the block-size logits. With categorical logits z^b and ordinal logits z^r, training minimizes

$$\mathcal{L} = \mathrm{CE}(z^b, b_t) + \frac{1}{83}\sum_{k=1}^{83} \mathrm{BCEWithLogits}\big(z^r_k, \mathbf{1}[r_t > k]\big). \tag{2}$$

We optimize Equation 2 with a model of hidden size 64, four heads, four layers, 0.1 dropout, and learning rate 5×10^{-4}; all inputs are z-score normalized with training statistics. Appendix B reports regression, random-forest, and architecture ablations.

3.3 Hybrid online scheduling

At block t, AdaBlock scans current logits for a high-confidence delimiter and returns the boundary b_t. The first block follows AdaBlock; later blocks pass the last eight realized tuples to the predictor and use only its budget r_t. Within the active block, masked positions are refined until the budget is reached. The decoder may then exit when all remaining predictions exceed confidence 0.8 or have stayed unchanged for two calls; a hard ceiling prevents unbounded refinement. Each initial proposal call and each refinement call counts toward NFE.

This hybrid follows directly from an observed feedback failure. Letting the predictor choose both controls often selects $(b_t, r_t) = (1, 1)$. The resulting trivial tuple makes another small prediction more likely, pushing the online feature distribution farther from the training corpus at every block. Preserving the reactive delimiter boundary breaks that loop. It also leaves a clean experimental question: does history forecast budget better than AdaBlock or a lookup table conditioned only on b_t?

4 Evidence protocol

Initial audited evaluation. The primary PAG comparison uses deterministic LLaDA-8B-Instruct decoding on all 1,319 GSM8K test problems and a 300-example index-stratified MATH-500 sample [Hendrycks et al., 2021]. GSM8K also includes a frozen development slice for choosing among eight scheduler variants and a disjoint 919-example confirmatory complement. The comparators are AdaBlock and a history-free lookup policy that returns the median training NFE for the current block size. Every method counts the boundary proposal call and all subsequent refinement calls. This corrects the legacy course result, whose PAG counter omitted the initial proposal and is excluded from every claim here.

Cross-model evaluation. Risk-calibrated PAG (RC-PAG) uses pinned LLaDA-8B-Instruct and Dream-v0-Instruct-7B revisions, temperature zero, generation length 256, at most 64 refinement calls, and block length at most 64. Estimator fitting, policy screening, and calibration use disjoint mixtures of GSM8K, MATH, and Mostly Basic Python Problems (MBPP) training data [Austin et al., 2021]. Confirmation uses 500 GSM8K test, 300 MATH-500, 100 sanitized MBPP, and 100 HumanEval prompts [Chen et al., 2021]; HumanEval is out of domain. Each candidate stop is compared with a same-state full-refinement shadow. A finite policy family is selected with exact binomial Learn-then-Test control at familywise level 0.05, but the certificate concerns this premature-commitment loss, not answer correctness.

Promotion gates and statistics. Results are paired by prompt. We report mean NFE, exact task accuracy, and latency, with 10,000-sample paired bootstrap intervals where available. A cross-model

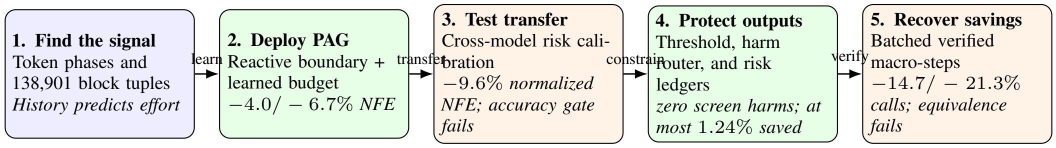


Figure 3: Approach progression. Each stage addresses the limitation exposed immediately before it; reported savings are relative to AdaBlock. The strongest corrected task result is PAG's 6.7% NFE reduction on MATH-500 with a 1.33-point accuracy decrease. Larger apparent reductions fail either task-quality or exact-equivalence gates; output-preserving screens save at most 1.24%.

Table 1: Corrected LLaDA evaluation. Accuracy is exact-answer percentage (higher is better); NFE counts all model forward calls (lower is better). The lookup baseline conditions only on block size.

Dataset	Method	Accuracy ↑	Mean NFE ↓	ΔNFE
GSM8K ($n = 1{,}319$)	AdaBlock	77.79	87.81	—
	size lookup	76.42	**82.14**	-6.5%
	PAG	76.19	84.26	-4.0%
MATH-500 ($n = 300$)	AdaBlock	38.00	119.65	—
	PAG	36.67	**111.67**	-6.7%

policy is promoted only if it reduces NFE on both models, beats the eligible history-free control, and has an accuracy-difference lower bound of at least -2 percentage points. Later protocols add model-wise minimum savings or exact sequence-equivalence gates. Partial screens and failed audits are reported as diagnostics, never as confirmatory wins.

5 Results

Figure 3 gives the experimental logic. We begin with the strongest corrected PAG result, then change one form of control at a time. A stage advances only when its evidence passes the corresponding accuracy, savings, or equivalence gate.

5.1 PAG reduces corrected NFE by up to 6.7%

Table 1 gives the corrected LLaDA result. Relative to AdaBlock, PAG saves 4.0% NFE on GSM8K and 6.7% on MATH-500, but loses 1.60 and 1.33 accuracy points. On the untouched GSM8K complement, the paired accuracy difference is -1.20 points with a 95% interval of $[-2.61, 0.22]$. The interval includes parity, but the point estimates do not support a dominance claim.

The size lookup is the decisive control: it uses less NFE than PAG on GSM8K and is slightly more accurate. History therefore does not explain the full gain. It does help forecast heterogeneous blocks offline, but much of the deployable signal is already captured by block size. The synchronized 150-prompt timing subset points in the same cautious direction: both decoders reach 84% accuracy and PAG is 0.31 seconds faster on average, but the paired latency interval crosses zero.

5.2 Risk calibration increases savings but exposes transfer failure

The cross-model experiment sharpens the distinction between prediction and control. Histogram-gradient-boosting risk estimators are strong on prompt-grouped holdouts: history features reach area under the receiver operating characteristic curve (AUROC)/Brier scores of 0.982/0.051 for LLaDA and 0.990/0.026 for Dream. All frozen low-risk candidates also satisfy their finite-sample shadow-loss tests. Yet Table 2 shows that this surrogate success does not transfer uniformly to answer accuracy.

Across both models and all four evaluation sets, normalized NFE falls from 1.000 to 0.904, compared with 0.968 for the best eligible nonlearned rule. This aggregate is not a positive headline: the predeclared accuracy-noninferiority gate fails, with a minimum lower confidence bound of -21.2 points. LLaDA remains close to parity but saves only 1–2%; Dream obtains larger compute re-

Table 2: Cross-model confirmation for the history policy. ΔAcc is the accuracy change in percentage points and ΔNFE is relative to AdaBlock.

Model	Dataset	AdaBlock Acc./NFE	RC-PAG Acc./NFE	ΔAcc	ΔNFE
LLaDA	GSM8K	78.2 / 86.89	78.2 / 85.06	$+0.0$	-2.1%
LLaDA	MATH-500	37.7 / 119.72	37.3 / 118.17	-0.4	-1.3%
Dream	GSM8K	69.8 / 103.04	53.6 / 89.46	-16.2	-13.2%
Dream	MATH-500	45.3 / 123.56	40.3 / 114.39	-5.0	-7.4%

ductions by changing too many answers. A well-calibrated same-state stop label is therefore an incomplete proxy for end-to-end quality under a new decoder family.

5.3 Conservative routing converges on output preservation

The next four protocols ask whether tighter routing can keep the savings without changing answers. V4 applies a local risk threshold and saves 9.1%/3.2% NFE on LLaDA/Dream, but Dream misses the predeclared 5% floor. V5 predicts both benefit and harm; harm AUROC is only 0.372/0.456, so the router is not promoted. V6 replaces local decisions with a prompt-level risk ledger, and v7 relaxes that gate. Both record zero harmful regressions in their screens, but save only 0.75%/0.45% and 1.24%/0.65%. This is the best output-preserving point reached by learned routing, but it gives back most of PAG's original efficiency gain. Protocols v2–v3 were superseded before GPU evidence was materialized and are not counted as experiments.

5.4 Verified macro-steps recover calls but not exactness

Under v8, a reference check evaluates a batch of fixed-depth proposals in one call. This lowers physical invocations from 83.47 to 71.19 on LLaDA and from 97.63 to 76.81 on Dream, reductions of 14.7% and 21.3%. These are the largest raw call reductions in the progression. They are not lossless results: 22 of 64 prompt trajectories disagree with AdaBlock. LLaDA latency also rises from 3.09 to 3.83 seconds because each invocation evaluates more rows. V9 audits the mechanism before another rollout. Execution fingerprints match. Successor states differ across depth-one and depth-two batch shapes. Figure 4 in the appendix shows this exactness test. The failed gate yields a final division of responsibility: learned histories propose budgets; a batch-invariant reference transition retains acceptance authority.

6 Discussion

Phase structure is useful but not sufficient. The trace analysis, tuple predictors, and risk estimators agree that refinement trajectories contain signal. The deployment results identify what that signal supports: forecasting relative effort. They do not support handing a learned score sole authority to commit a block. Current-state boundary and stability rules use contemporaneous evidence [Lu et al., 2026, Mohamed et al., 2025, 2026]; the problematic transfer here occurs when history predicts a future commitment. This difference explains why high risk-model AUROC can coexist with poor Dream accuracy and why reducing the score threshold can preserve outputs while eliminating the compute benefit.

History-free controls should be mandatory. Block size encodes both span length and AdaBlock's semantic-boundary decision. A lookup table on that single variable beats the learned scheduler on GSM8K, so comparisons against a fixed budget or AdaBlock alone overstate the value of history. Future learned schedulers should predict a bounded residual over a strong current-state prior and show that history improves a paired frontier, rather than reporting only an absolute saving.

Verification is an implementation property. The speculative pilots suggest that batching reference transitions could reduce sequential depth, but the failed equivalence audits prevent a lossless claim. Exactness requires more than matching a draft token with a batched logit row: the reference operator, cache state, attention mask, block boundary, and batch shape must induce the same complete successor as serial decoding. This batch-invariance audit complements token-level verification in self-speculative decoders [Gao et al., 2025, Han et al., 2026]. Physical model calls, evaluated rows, latency, and memory are separate costs and should be reported separately.

7 Limitations

The study covers two 7–8B model families, 256-token generations, and math/code benchmarks; it does not show that phase history is unhelpful for larger models, longer contexts, or a scheduler trained jointly with the model. CPD evidence is qualitative and detector-dependent. The risk certificates apply to a declared shadow loss and calibration mixture, not semantic correctness or arbitrary deployment data. Code-task exact-match rates are sensitive to extraction and test harnesses, so they appear in the appendix rather than drive the main claim. Latency was measured at batch size one on a single-GPU workflow and is not a serving benchmark. Finally, the v8–v9 results diagnose this implementation of batched verification; they do not rule out verified acceleration with stricter batch-invariant kernels or serial verification.

8 Conclusion

PAG began with the hypothesis that reasoning phases should receive different refinement budgets. The hypothesis is partly right: token stabilization is structured, block histories predict compute, and a hybrid scheduler reduces LLaDA model calls on GSM8K and MATH-500. The stronger claim does not survive the full evidence. A block-size lookup is competitive, cross-model risk control fails task-accuracy transfer, conservative gates save little, and speculative batching fails exactness audits.

Our experiments support a separation of responsibilities. Learned phase models are well suited to proposing how much capacity to allocate. Semantic boundaries should remain tied to current model evidence, and token acceptance should remain with an audited reference transition. A next system should therefore combine a bounded history residual with batch-invariant verification and treat an unsuccessful equivalence or task-quality gate as a normal fallback, not a result to average away.

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A Verification equivalence audit

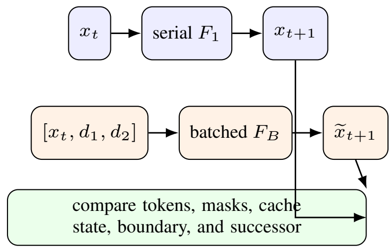

Figure 4: Exactness audit for verified macro-steps. A batched row is eligible only when it induces the same complete successor as the serial reference transition. The v8 pilot disagrees on 22 of 64 prompts, and v9 rejects the depth-dependent batch shape before rollout.

B Predictor and integration diagnostics

Continuous regression. The first Transformer predicts real-valued controls with normalized validation loss 0.096. Deployed on 30 arithmetic prompts, it reaches 73% accuracy versus 93% for AdaBlock and averages 8.1 blocks rather than 12.3. Inspection shows interpolated block sizes merging semantic spans and producing repeated or truncated text. This run motivated the categorical block head; it is not used as a baseline in the audited evaluation.

Random forest. A 200-tree, depth-15 multi-output forest uses rolling statistics from a five-tuple window. On 90,994 training and 22,907 held-out examples, block-size mean absolute error (MAE) is 7.24 and stabilization-step MAE is 1.66. The last block size is the largest individual feature (importance 0.214), supporting the explicit size-lookup control.

Transformer ablation. The categorical/ordinal sweep varies context length, hidden size, attention heads, layers, dropout, and learning rate. The selected $(w, d, h, l) = (8, 64, 4, 4)$ model reaches validation loss 2.216957. Wider models do not improve this objective: at four layers, the best losses for widths 32, 64, 128, 256, and 512 are 2.233, 2.217, 2.238, 2.289, and 2.336.

C Complete cross-model confirmation

Table 3 reports every dataset/model pair from the v1 confirmation; the code tasks are retained for completeness but do not drive the main claim.

Table 3: Complete v1 confirmation. Accuracy is percentage and NFE is the mean number of physical model calls. HumanEval is declared out of domain.

Model	Dataset	Method	Accuracy	Mean NFE
LLaDA	GSM8K	AdaBlock / RC-PAG	78.2 / 78.2	86.89 / 85.06
LLaDA	MATH-500	AdaBlock / RC-PAG	37.7 / 37.3	119.72 / 118.17
LLaDA	MBPP	AdaBlock / RC-PAG	4.0 / 4.0	71.66 / 70.57
LLaDA	HumanEval	AdaBlock / RC-PAG	32.0 / 33.0	95.48 / 94.39
Dream	GSM8K	AdaBlock / RC-PAG	69.8 / 53.6	103.04 / 89.46
Dream	MATH-500	AdaBlock / RC-PAG	45.3 / 40.3	123.56 / 114.39
Dream	MBPP	AdaBlock / RC-PAG	3.0 / 0.0	80.02 / 54.25
Dream	HumanEval	AdaBlock / RC-PAG	48.0 / 16.0	84.78 / 64.50

D Artifact-backed protocol ledger

Table 4 records the evidence source for each reported phase. Hashes are shortened only for readability. All records are non-mock and pin model and dataset revisions in their environment manifests.

Table 4: Artifact provenance for numerical claims.

Evidence	Config/run hash	Authoritative artifact
Initial PAG	Strategy 1	regraded report summarized in final report
RC-PAG v1	`5197d981c0bf`	`report/summary.json`, `claim_audit.json`
Local risk v4	`8cce1af8e35b`	`screening_summary.json`
Advantage v5	`7688c5235bd4`	screen records, `advantage_manifest.json`
Risk ledger v6	`58e76c3eb9b1`	`readiness_audit.json`
Relaxed gate v7	`c1eda289fb08`	`readiness_audit.json`
Verified pilot v8	`d36b982c2388`	`parity_audit.json`
Equivalence audit v9	`588204cfb482`	`equivalence_audit.json`

E Accounting and reporting rules

NFE denotes physical invocations of the model forward function, including the initial proposal used to choose a boundary. A batched invocation counts as one physical NFE and separately contributes one evaluated row per batch element. We do not translate NFE reductions into FLOP, energy, or latency claims. Accuracy is paired on identical prompt IDs. A run with a failed manifest, incomplete coverage, a mock record, or a failed claim gate is ineligible for a positive aggregate.

F Reproducibility, compute, and assets

Configurations and statistics. The anonymous supplement contains the implementation, exact YAML configurations, pinned revisions, run manifests, and artifact summaries referenced in Table 4. The initial audit uses seed 20260710; the cross-model protocols use seed 20260729. Confidence intervals are percentile intervals from 10,000 paired bootstrap resamples of prompt IDs. The risk tests use exact binomial bounds at familywise level 0.05. The supplement documents the commands for the initial Strategy 1 evaluation and each RC-PAG protocol. Raw checkpoints and benchmark copies are not redistributed; the loaders fetch the pinned public assets.

Compute. We estimate approximately 160 A100-equivalent GPU-hours for the completed experiments. Trace collection and the original PAG study account for about 40 hours; the six materialized protocols v1 and v4–v8 account for about 20 hours each, or 120 hours. V2–v3 were superseded before full GPU runs, and v9 stopped at the equivalence preflight before a benchmark rollout; incomplete job attempts and CPU-only analysis are excluded. The estimate assumes one NVIDIA A100 PCIe (80 GB), with LLaDA and Dream loaded one at a time in bfloat16. The recorded environment used Python 3.11.15, PyTorch 2.5.1 with CUDA 12.1, Transformers 4.46.3, Datasets 4.8.5, scikit-learn 1.9.0, and math-verify 0.9.0. Atomic records retain per-prompt elapsed time; the v8 pilots ranged from roughly one to six seconds per method/prompt.

Licenses. The pinned LLaDA checkpoint is MIT licensed and the pinned Dream checkpoint is Apache-2.0 licensed. GSM8K, MATH-500, and HumanEval are used under their MIT licenses; MBPP data are used under CC BY 4.0. We cite each model and benchmark at first use and preserve their original terms. The supplement contains no model weights or benchmark examples beyond the aggregate statistics and one short diagnostic trace shown in Figure 2.

G Broader impact

Reducing sequential model calls can lower latency and energy when the extra batch work does not erase the gain. It can also make generative systems cheaper to deploy at scale. The experiments expose the corresponding reliability risk: an efficiency controller that commits tokens too early can retain attractive proxy metrics while changing answers substantially. Systems using learned scheduling should therefore keep a reference fallback, audit task quality, and report batch work as well as call count. This work releases no new foundation model or scraped dataset and does not expand the underlying models' capabilities; any downstream misuse or bias remains that of the base models and their deployment context.

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